Jan 27, 17:30

Books:

JP Katoen: Principles of model checking.

$$\text{Model} \longleftrightarrow \text{Requirement}$$

M

R_1

R_2

$\vdots$

R_n

$\boxed{M \models R_1}$ $\hookrightarrow$ model M satisfies the requirement R_1

$$\begin{tikzcd} M \arrow[dr] & & \phi_m \arrow[dl] \\ & \boxed{MC} \arrow[dl] \arrow[dr] & \\ \text{Yes} & & \text{No : counter example} \end{tikzcd}$$

$\mathbb{P}$ → program

x, y, z

init:

$x = 0$

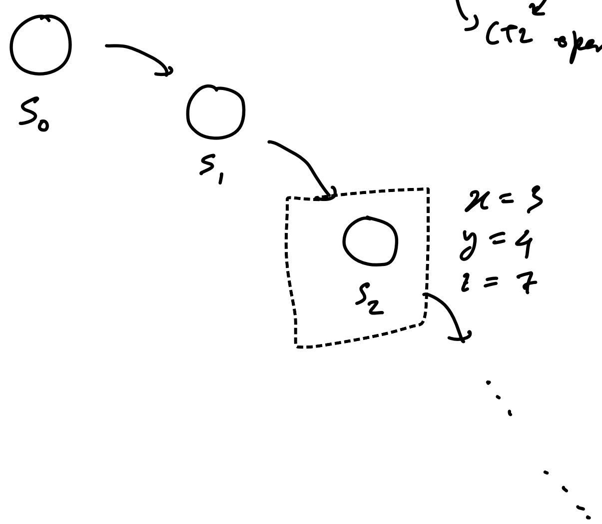
$y = -1$

$z = -2$

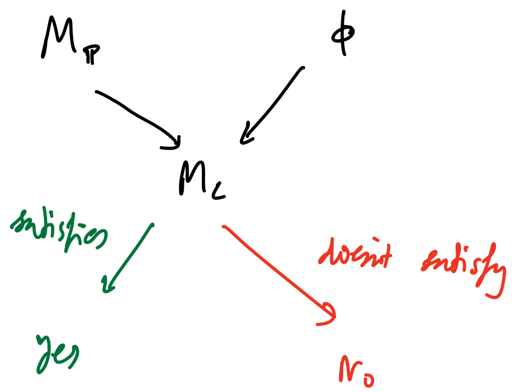
Eventually, $4 \le y \le 10$

AF (AG $4 \le y \le 10$)

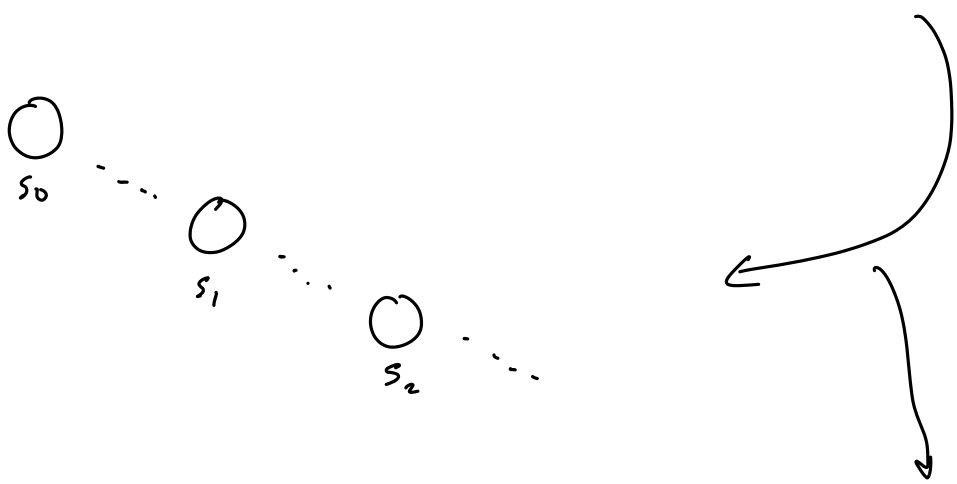
CTL operator



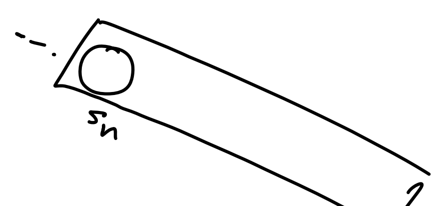
ϕ : situation where $y \le 10$



→ gives a counter evidence



shows a path where $y > 10$



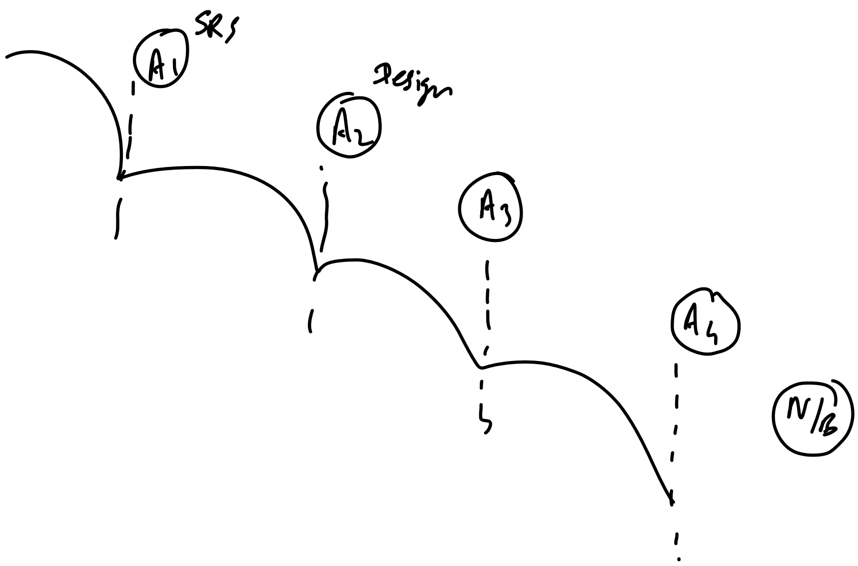
Model Checking is

1. completely automatic

2. deductive

→ Theorem proving

Waterfall model



Verification vs Validation

$A\,F\,\phi$	A → for all paths
$E\,F\,\phi$	E → for existing ? paths